

Unit – I	Introduction to DBMS and ER Data Model: Data Storage: File processing system, Disadvantages; DBMS: Need of DBMS, Terms: Data, Database, Metadata, Data Dictionary, Database System, Database Management System, Data Abstraction, Data Independence, System Architecture of DBMS, Data Model: Definition, ER and Relational Data Model, Object Oriented, Object Relational Models; ER Model: Entity, Entity Set, Attributes, Primary Key, Relationship, Types and Attributes of Relationship, Role, Cardinality Ratio, Participation Constraint, Weak Entity Set, EER Features
Unit – II	Relational Data Model: Terms: Relation, Schema, Attributes, Tuples, Domains, Relation Degree (or Arity) and Cardinality, Relation Intention and Extension, Super Key, Candidate Key, Primary Key and Foreign Key, Relational Model Constraints, Schema Diagram, ER to Relation Mapping, Detailed storage architecture.
Unit – III	Relational Algebra and Relational Calculus: Relational Algebra and its Operations: Set Theoretic Operators (Union, Intersection, Cartesian product and Division), Relational Algebra operators (Projection, Selection, Join and Rename), Relational Calculus: TRC, DRC, Database Language: SQL (DDL, DML, DCL), QBE.
Unit – IV	Normalization: Normalization: Anomalies of un-Normalized Relation, Need of Normalization, Pros and Cons of Normalization, Functional Dependency: Trivial, Full, Partial, Transitive, Multi-valued, Join, Inclusion Dependency, Dependency Diagram, Inference Rules for Functional Dependencies, Closure of Functional Dependencies, Algorithms to find: 1. Candidate Key, 2. Closure of Attribute Set, 3. Minimal Cover of Functional Dependencies, Normal Forms: Checking of Lossless Join Decomposition and Dependency Preservation, Normal Forms: 1NF, 2NF, 3NF, BCNF, 4NF, 5NF
Unit – V	Transaction Processing & Management: Transaction: Concept, ACID properties, Transaction States; Schedule: Definition, Types of Schedule, Serializability, Conflict and View Serializability, Precedence Graph, Recoverable Schedule, Cascade less Schedule, Deadlock, Concurrency Control Protocols: Lock Based, Timestamp Based Protocol, Recovery System: Log based Recovery, Checkpoint, Shadow Paging